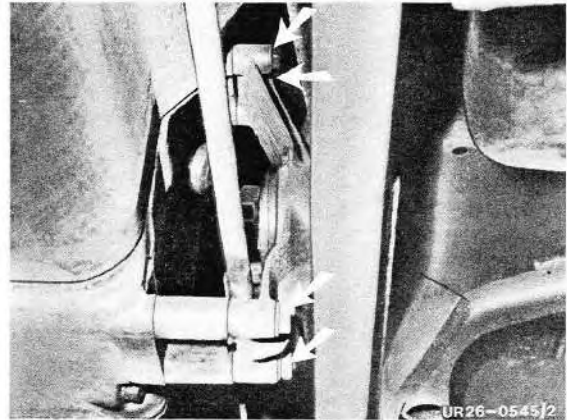


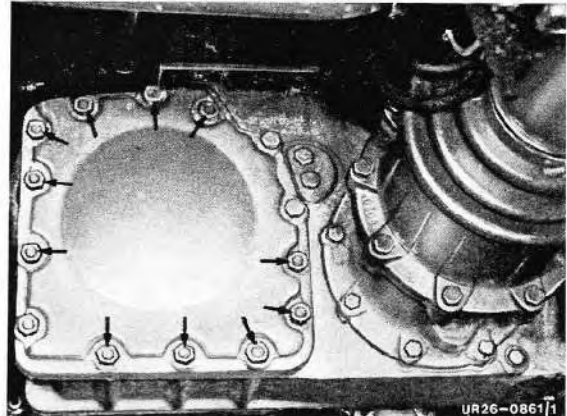
717.9
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- 9 Support shift and planetary gear and unscrew transmission mount at the right.



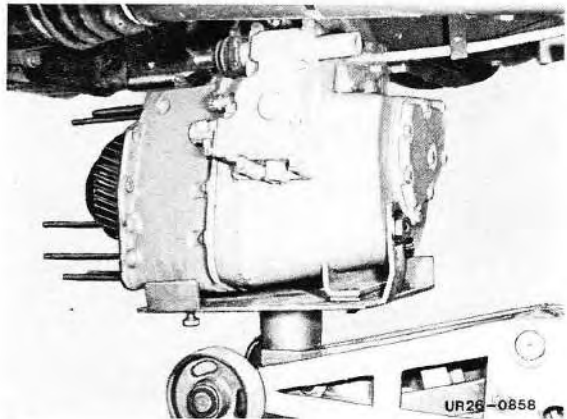
- 10 Remove shift and planetary gear.

Note: One bolt is located on the front of the transfer case.



- 11 Draw back shift and planetary gear, lower and remove.

- 12 Remove gasket and clean sealing surface.



Installation, Transfer Box Installed

- 1 Attach gasket, introduce shift and planetary gear and raise.

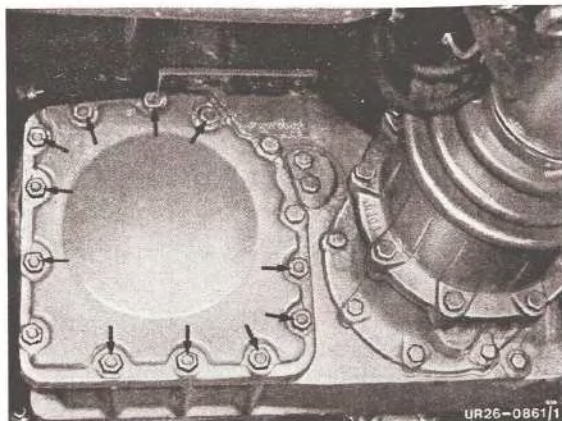
- 2 Attach transmission mount at right and tighten. Tightening torques, see 1.7/1.

Note: Use fastening bolts only once.



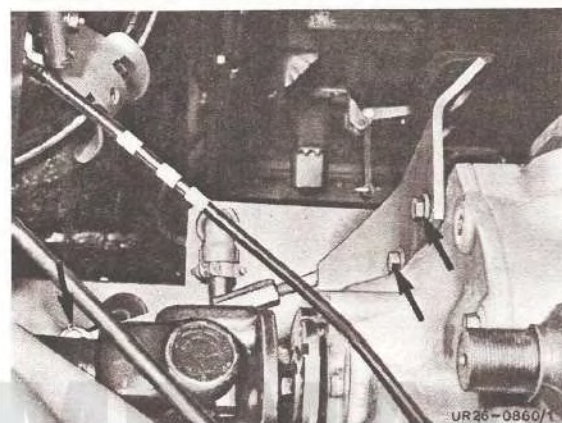
26.13 Removal and Installation of Main Transmission

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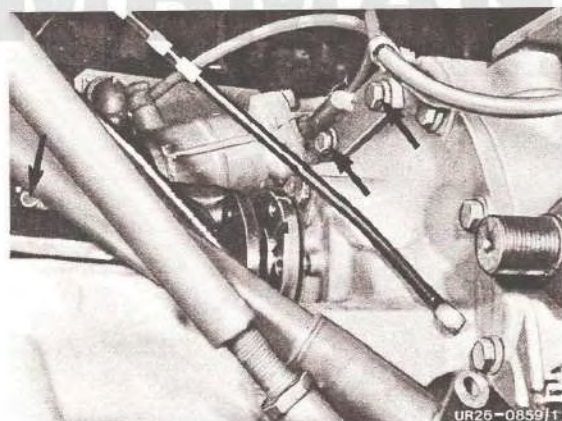


3 Bolt on shift and planetary gear.

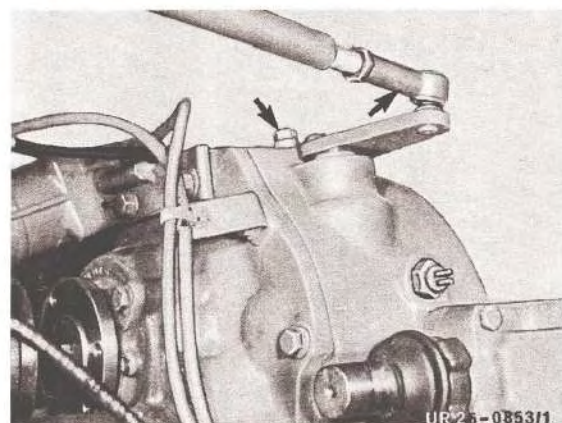
Note: One bolt is located on the front of the transfer case.



4 Detach special tool No. 7.



5 Adjust pneumatic cylinder, fit and secure piston rod.
See 1.5/1 for adjustment.



6 Attach transmission breather and forward/reverse
shift linkage.

7 Attach gear shift linkage, speedometer shaft with sealing sleeve, shift gate indicator, switch for reversing lamp, switch-over valve line and pneumatic cylinder, secure speedometer shaft with lead seal.

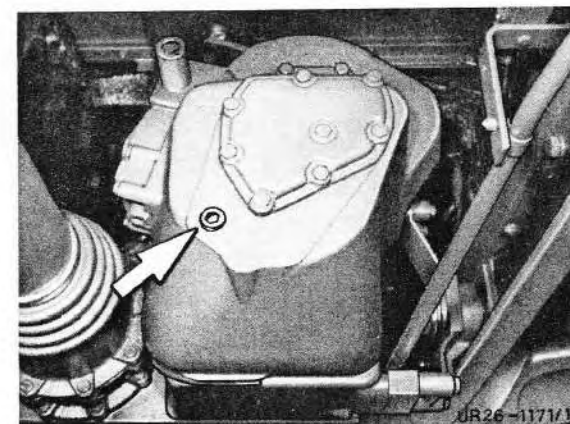
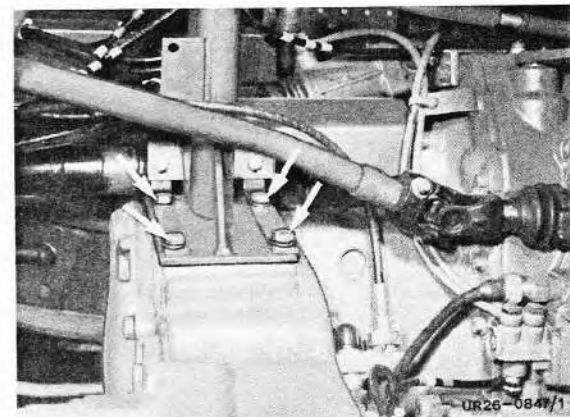
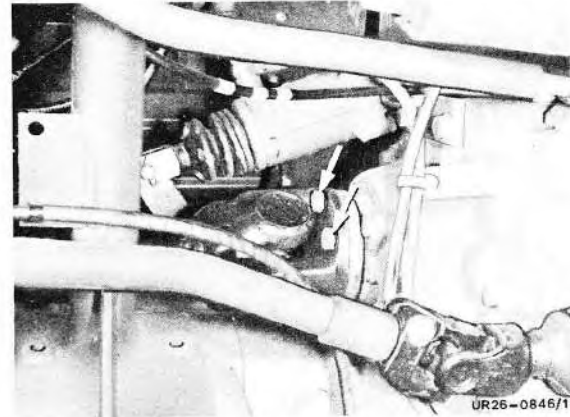
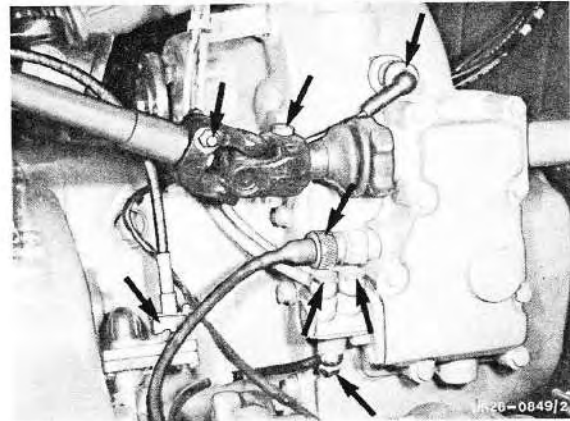
8 Adjust main shift lever, see 12.1/ 2.

Note: From chassis end No. 108 852 the sealing sleeve 303 9972240 is installed as standard.

9 Attach input shaft.

10 Attach cover plate and safety support.

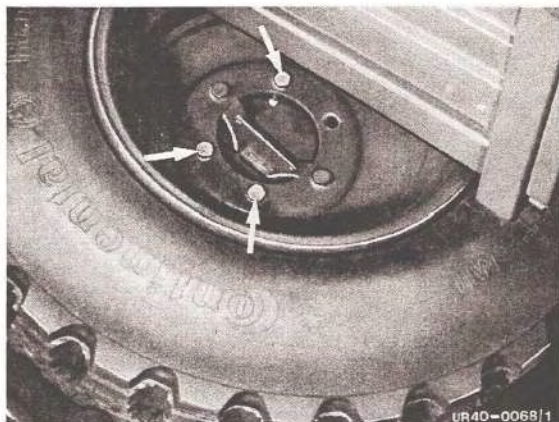
11 Fill main transmission with oil.
For capacity, see 1.2/1.



26.13 Removal and Installation of Main Transmission

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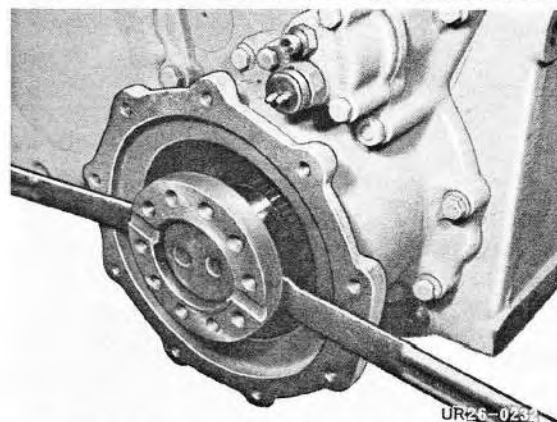
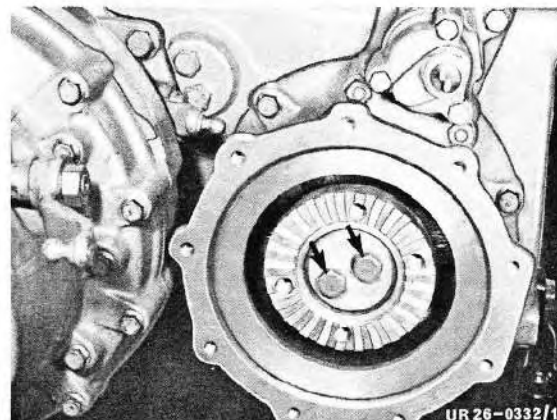


12 Fit spare wheel.

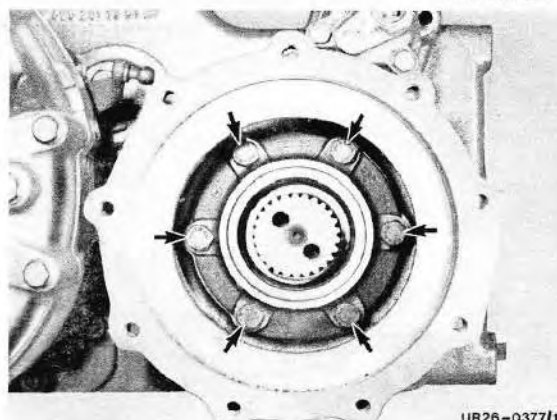
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Removing front axle PTO

- 1 Clamp main transmission in trestle.
- 2 Lock front axle PTO and remove supporting disk.
- 3 Force off flange.

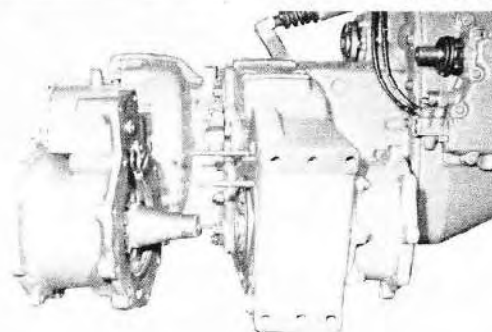
**Up to transmission No. 013 286**

- 4 Remove bearing cover from intermediate housing.
- 5 Remove shaft sealing ring from bearing cover.

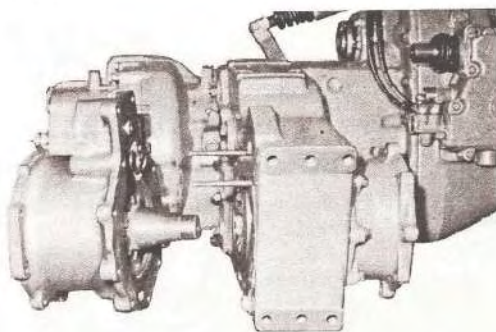


- 6 Remove intermediate housing.

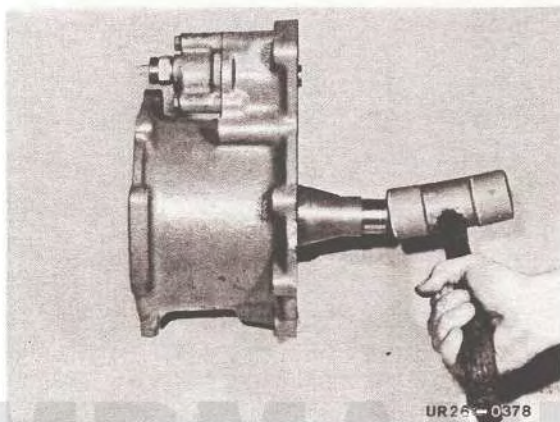
Note: Pay attention to shims and compression spring.



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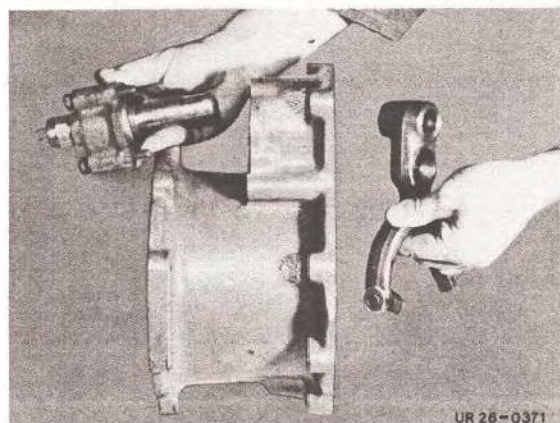
UR26-1183



UR26-0378



UR26-0370



UR 26-0371

From transmission No. 013 287

- 7 Remove intermediate housing.

Note: Pay attention to shims and compression spring.

Disassembling front axle PTO

up to transmission No. 013 286

- 1 Remove PTO shaft from intermediate housing.
- 2 Pull deep-grooved ball bearing off PTO shaft.

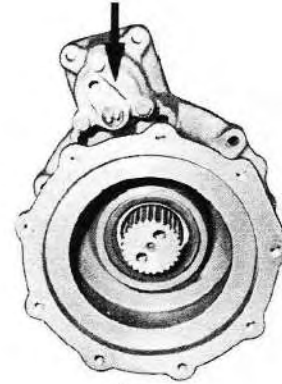
- 3 Remove shift dog.

- 4 Remove shift cylinder and shift fork with fulcrum pads for four-wheel drive.

- 5 Clean and check all parts, exchanging if necessary.

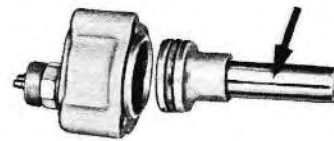
From transmission No. 013 287

- 6 Remove shift cylinder.



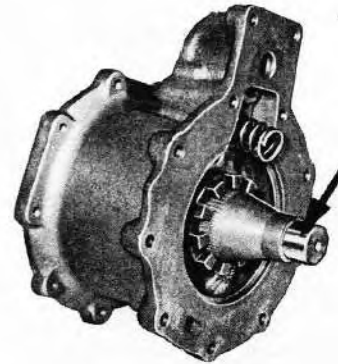
UR26-1180/1

- 7 Remove shift piston.



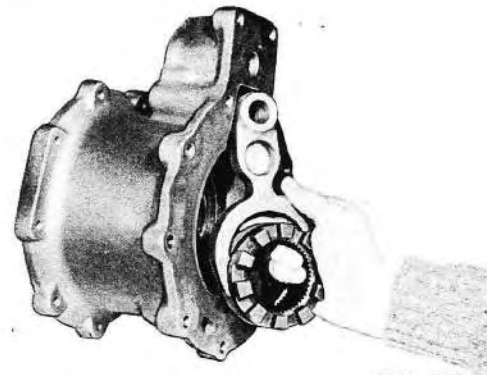
UR26-1182/1

- 8 Remove PTO shaft and take off compression spring.



UR26-1179/1

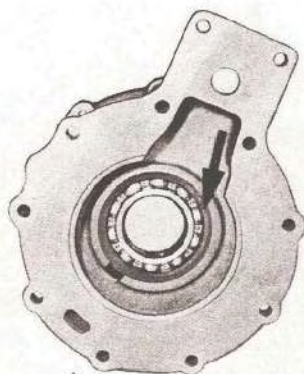
- 9 Remove shift dog and shift fork.



UR26-1178

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UR26-1181/1

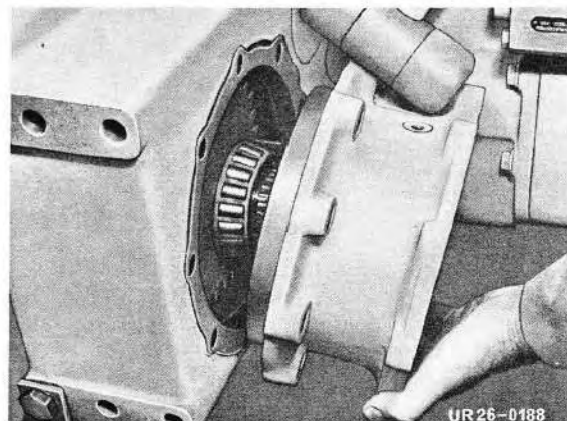
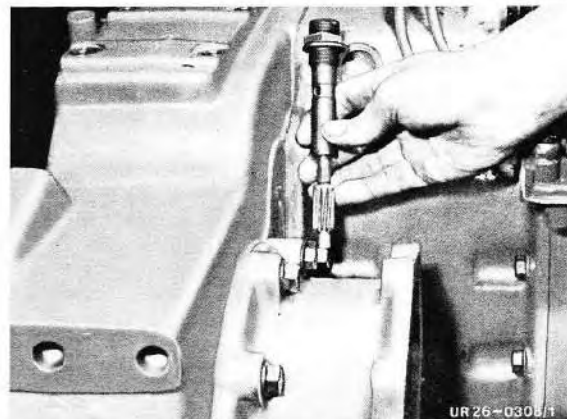
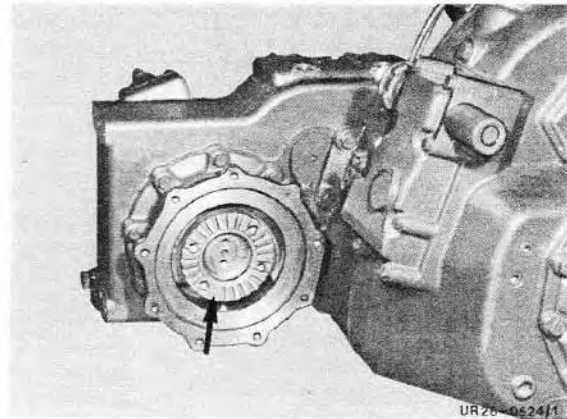
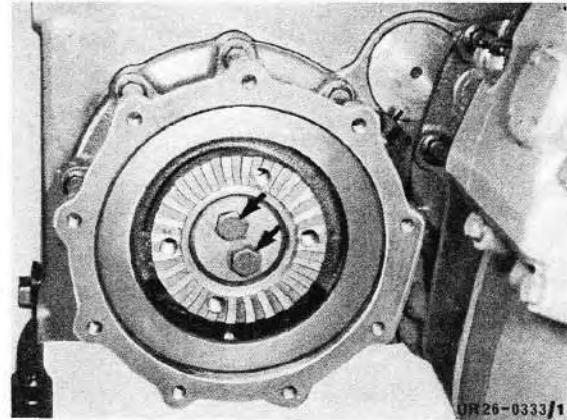
10 Release deep-groove ball bearing and remove.

11 Clean and check all parts, exchanging if necessary.

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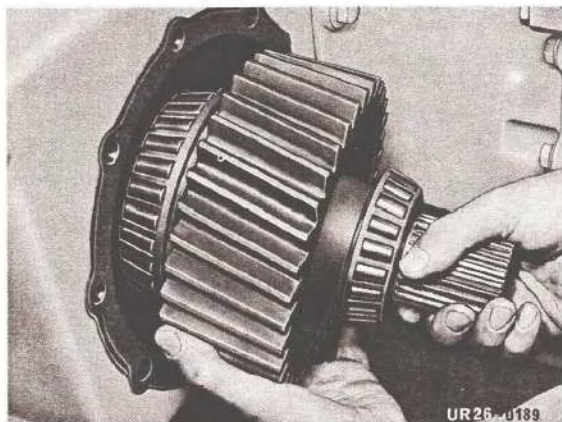
Removal of rear axle PTO

- 1 Clamp main transmission in trestle.
- 2 Lock rear axle take-off and remove supporting disk.
- 3 Force flange off.
- 4 Remove speedometer drive.
- 5 Remove intermediate housing, taking out shaft seal and speedometer drive gear.

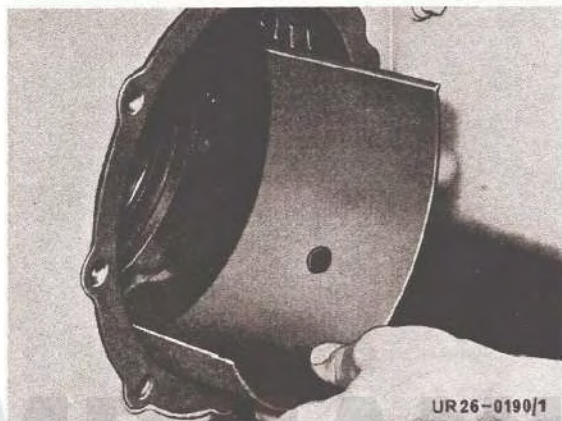


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- 6 Take output gear out of transfer box.



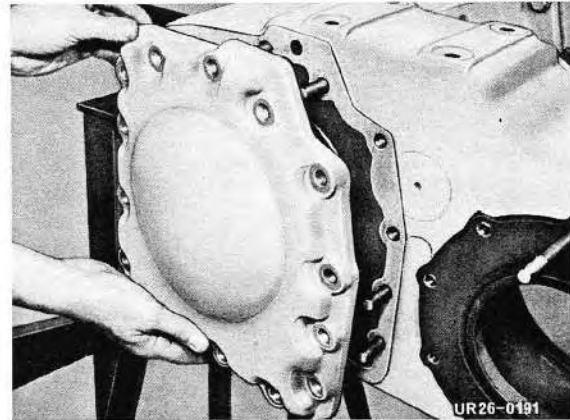
- 7 Take oil baffle out of transfer box.
- 8 Pull taper roller bearing off PTO gear.
- 9 Clean and check all parts, exchanging if necessary.

Note: An oil baffle with bore is installed as from transmission No. 000 360.

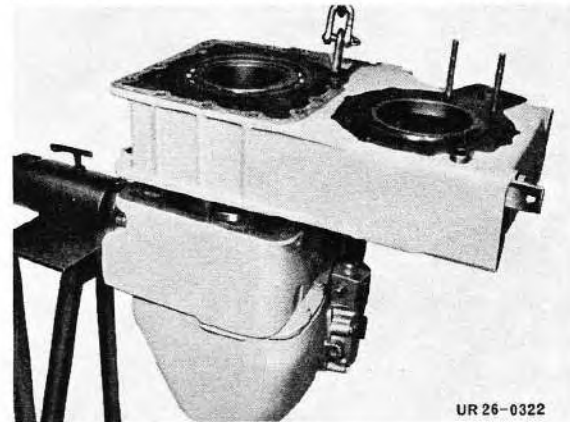
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Removing transfer box

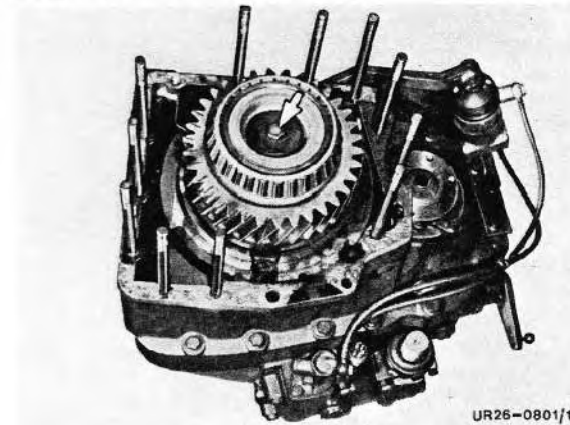
- 1 Clamp main transmission in trestle.
- 2 Remove intermediate housing, referring to 3.1/1.
- 3 Remove bearing cover.
Disassemble of front axle — and rear axle take-off, referring to 3.1/1 and 3.2/1.



- 4 Turn main transmission by 90°.
- 5 Take out connecting bolts.
- 6 Remove transfer box housing.

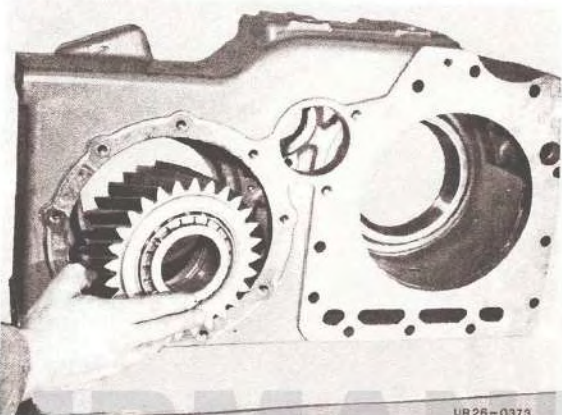
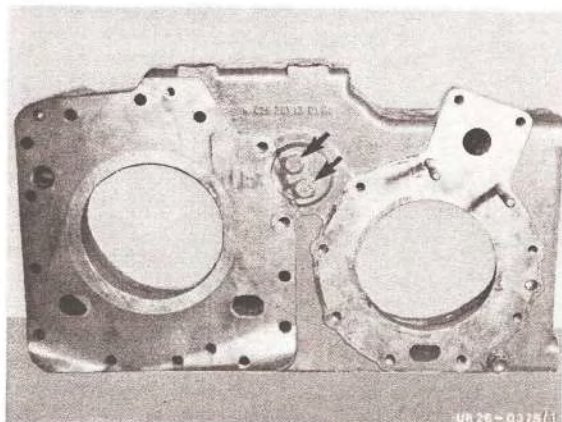


- 7 Remove supporting disk and take drive gear out.
- 8 Pull taper roller bearing off output gear.



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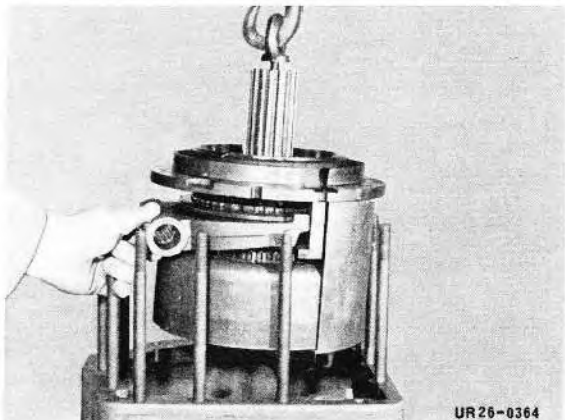
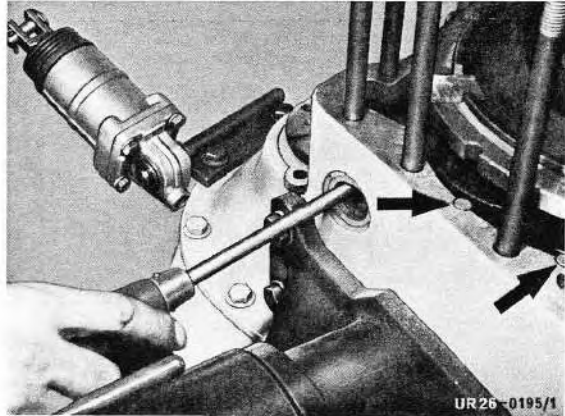
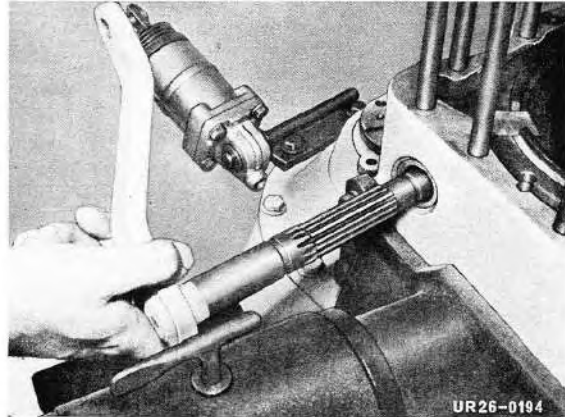
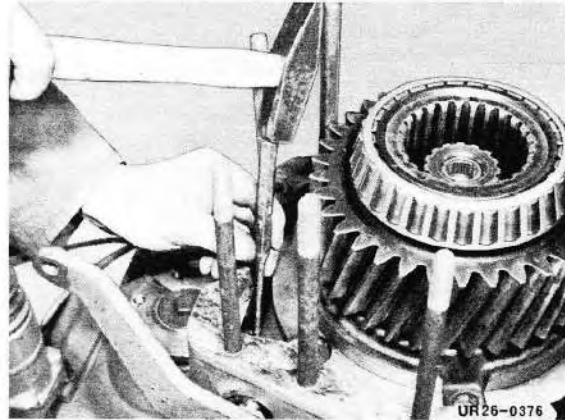


Disassembling transfer box

- 1 Detach transfer box, referring to 3.3/1.
- 2 Remove retaining bolts with supporting disk.
- 3 Pull off idler gear shaft and remove idler gear.
- 4 Remove outer race of taper roller bearing.
- 5 Remove spacing sleeve and shim.
- 6 Clean and check all parts, exchanging if necessary.

Removing planetary gear

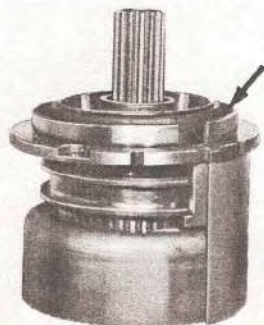
- 1 Detach transfer box, referring to 3.3/1.
- 2 Remove shift cylinder from shift shaft.
- 3 Knock clamping pin out of selector fork.
- 4 Remove shift shaft.
- 5 To remove selector fork, lift up using screwdriver. Remove locking pins and released clamping pin.
- 6 Take out planetary gearset with selector fork.
- 7 Remove shaft seal.



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Disassembly of planetary gear

- 1 Remove synchronizer.



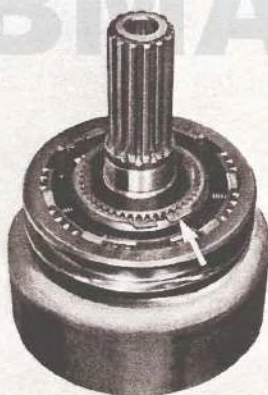
UR 26-0804/1

- 2 Remove shim and spacing tube.



UR 26-0805

- 3 Release and remove synchromash member.
- 4 Disassemble synchromash member, referring to 7.1/1.



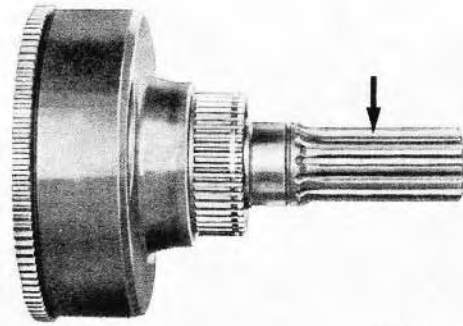
UR 26-0806/1

- 5 Release synchronizing dome and knock off downwards.



UR 26-0386

- 6 Remove planet-pinion carrier from ring gear.



UR 26-0807/1

- 7 Release and remove deep-groove ball bearing.



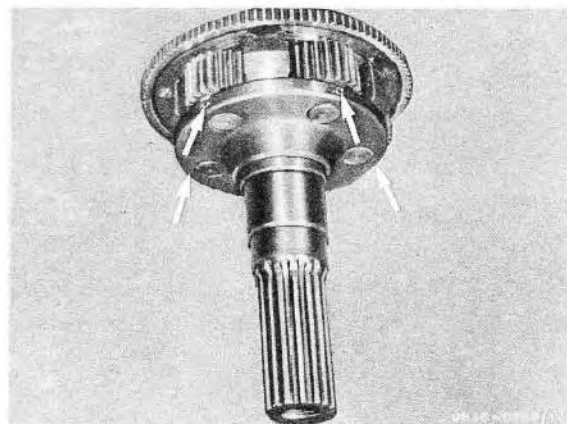
UR 26-0808/1

- 8 Release and disassemble ring gear. Valid up to transmission No. 012 581.

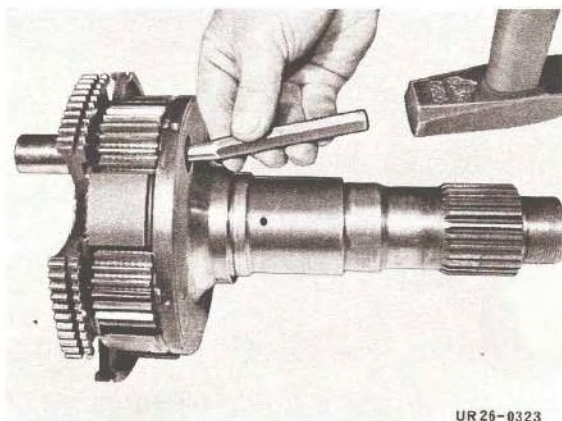


UR 26-0809

- 9 Release planet-pinion carrier.



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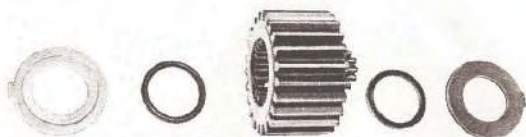


- 10 Take planet-pinion shaft out.

- 11 Remove bearing needles and shims.

Note: Bearing needles in planet-pinion must always be exchanged in sets (22 per set).

- 12 Clean and check all parts, exchanging if necessary.



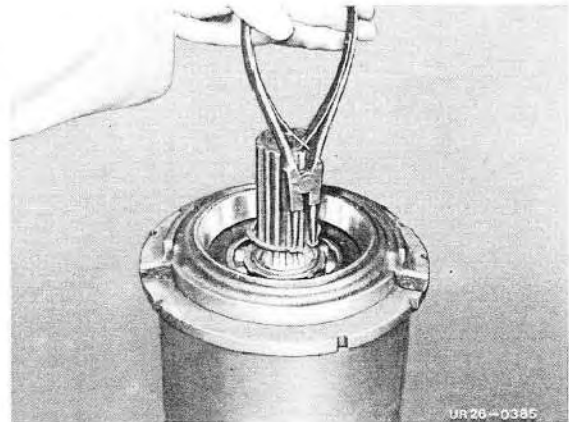
UR 26-0324

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SA 35 977

Disassembly of Planetary Gear

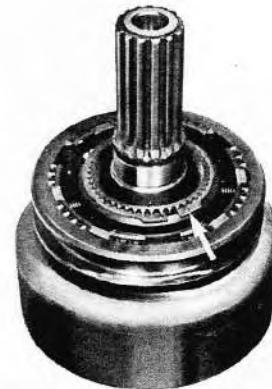
- 1 Remove planetary gear, see 3.4/1.
- 2 Remove retaining ring and shim.
- 3 Draw off synchronizer body together with deep-groove ball bearing.
- 4 Remove synchronizer ring and draw off deep-groove ball bearing in synchronizer body.



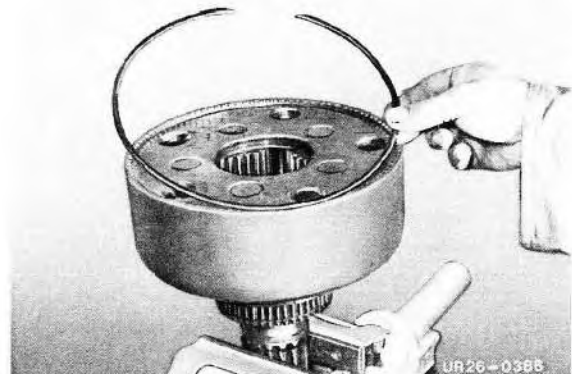
- 5 Remove shim and spacing tube.



- 6 Release and remove synchromesh member.
- 7 Disassembly synchromesh member, referring to 7.1/1.

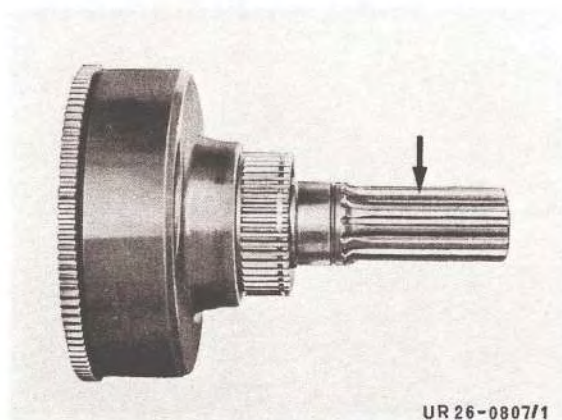


- 8 Release synchronizing dome and remove from below.



26.13 Disassembly of Main Transmission

717.900
SA 35 977



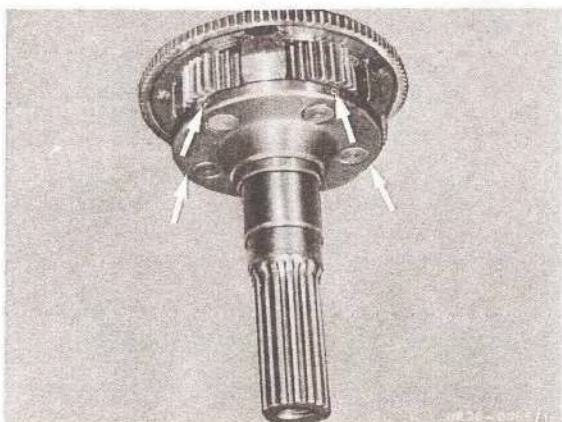
9 Remove planet pinion carrier from ring gear.



10 Release and remove deep-groove ball bearing.



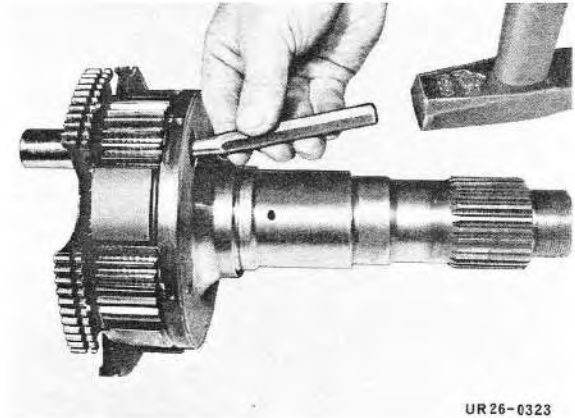
11 Release and remove ring gear.
Valid to transmission No. 012 581.



12 Release planet pinions.

717.9
SA 35 977

- 13 Remove planet gear shaft.

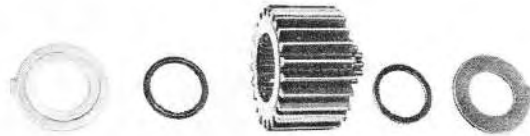


UR 26-0323

- 14 Remove bearing needles and distance pieces.

Note: Bearing needles must only be changed in sets (22 in each) in the planet gear.

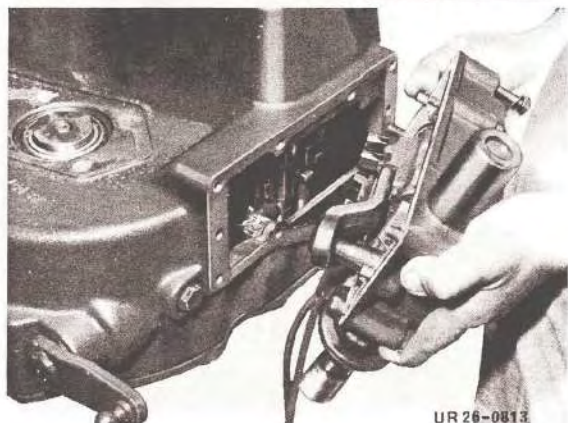
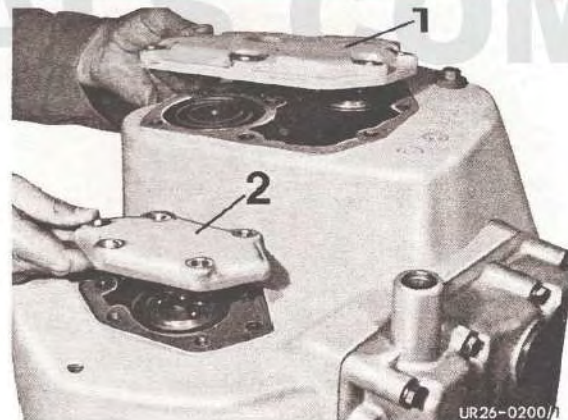
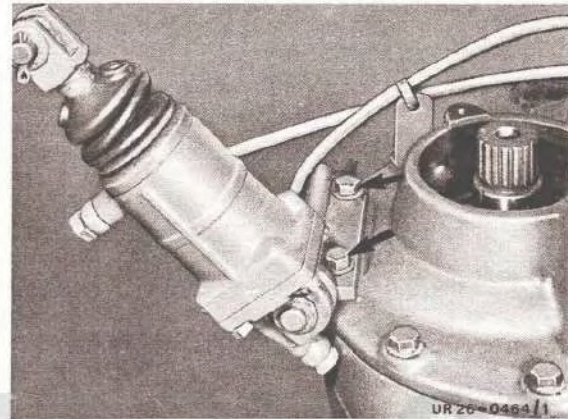
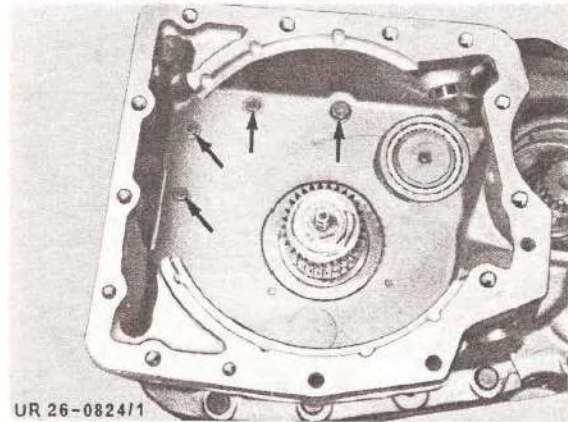
- 15 Clean all parts, check and replace if necessary.



UR 26-0324

Disassembly of transmission

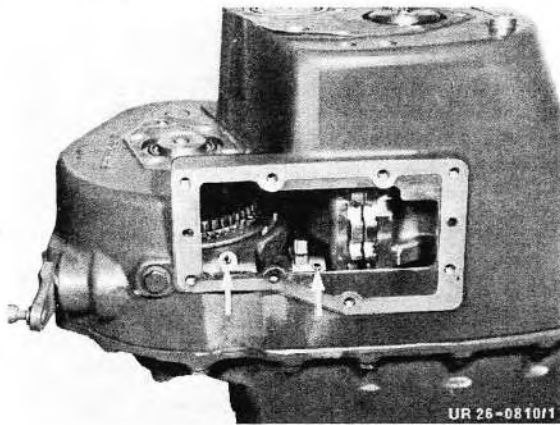
- 1 Remove oil pump, referring to 11:1/1.
- 2 Remove fastening bolts from planetary gear casing.
- 3 Detach pneumatic cylinder with bracket.
- 4 Turn transmission by 180°.
- 5 Remove bearing covers for main shaft/counter-shaft (1) and for input shaft (2).
- 6 Detach shift cover.



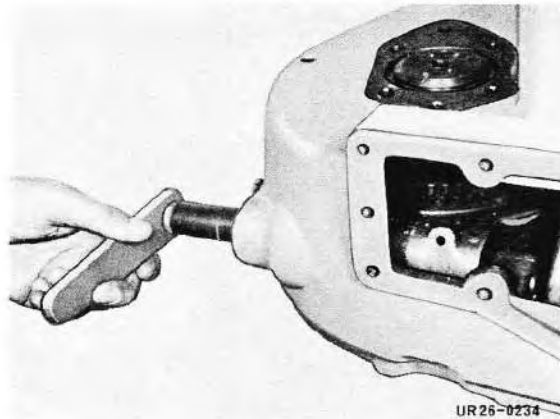
26.13 Disassembling main transmission

717.9

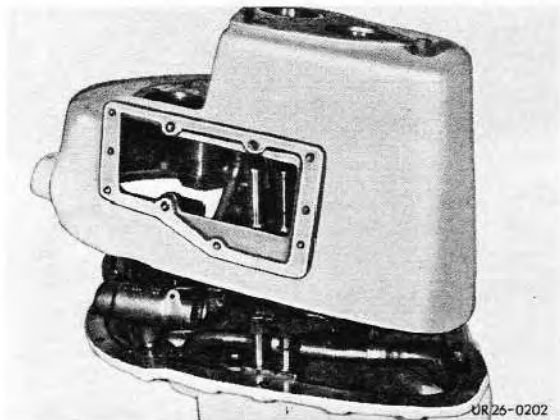
SA 35 977



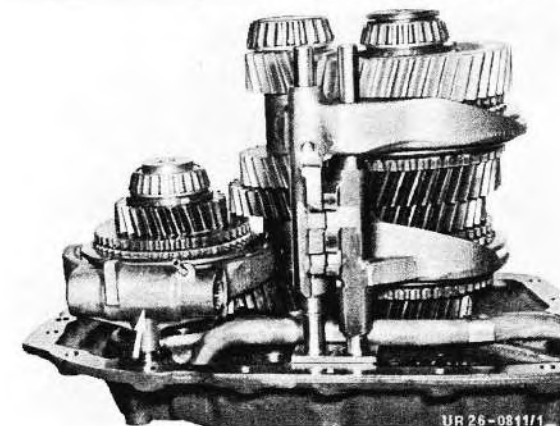
- 7 Move selector fork to reverse speed position, knocking out clamping pin at forward/reverse selector fork and reverse shift lock.



- 8 Remove shift shaft and reverse shift lock.
- 9 Remove remaining bolts from main transmission.

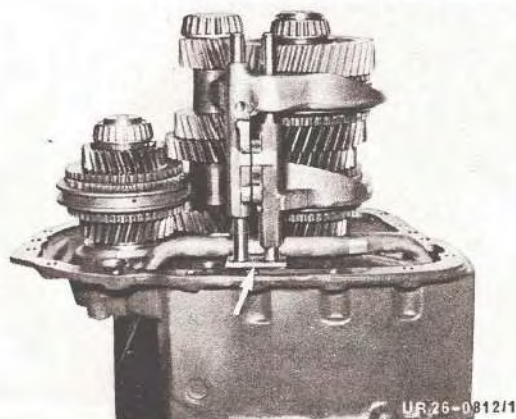


- 10 Attach fixture, lift up selector forks with tire lever and remove housing.

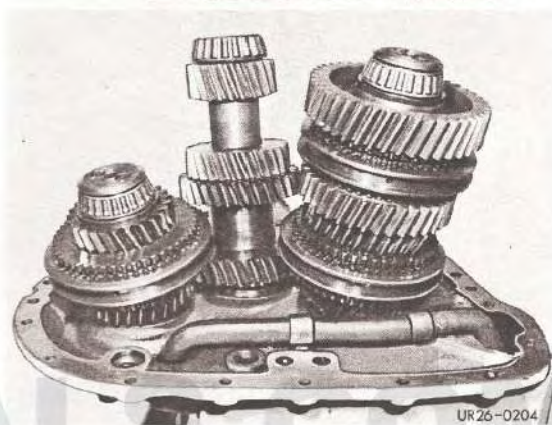


- 11 Remove selector fork for forward/reverse speed.

12 Press shifting forks upwards, remove shifting shafts and guide plate.



13 Move main shaft, counter shaft and input shaft to one side.

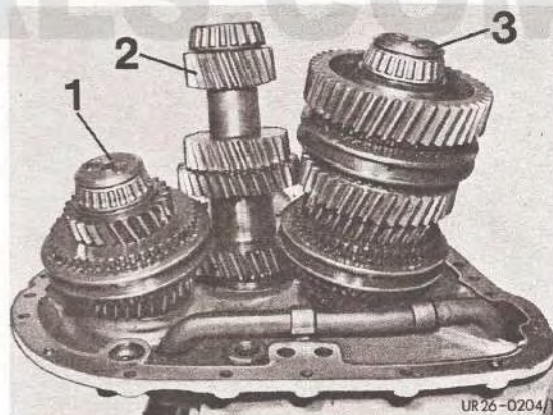


14 Remove input shaft (1), counter shaft (2) and main shaft (3).

15 Disassemble input shaft, see 5.1/1.

16 Disassemble main shaft, see 6.1/1.

17 Turn transmission through 90°.

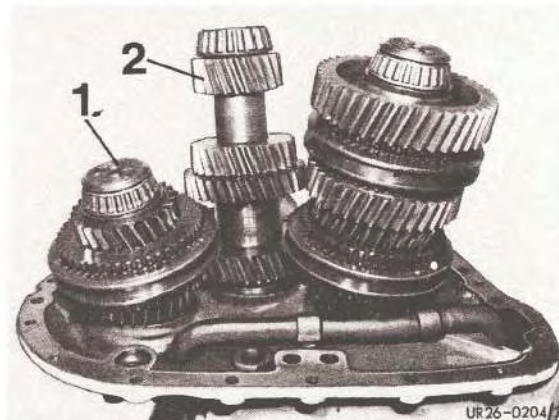


Valid for Transmission Model 717.900

18 Remove input shaft (1), and counter shaft (2).

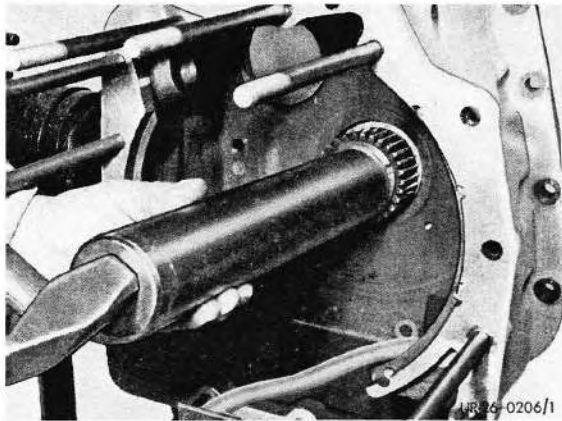
19 Disassemble input shaft, see 5.1/1.

20 Turn transmission through 90°.



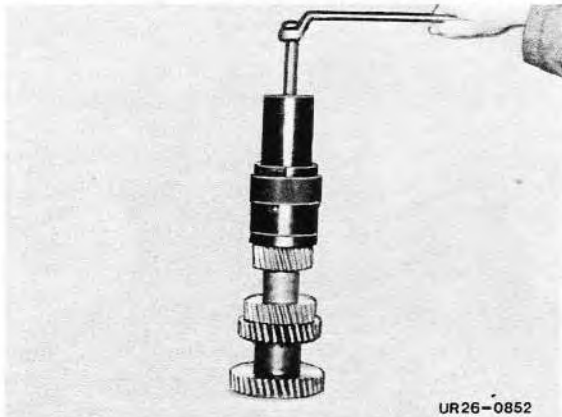
26.13 Disassembly of Main Transmission

717.9
SA 35 977



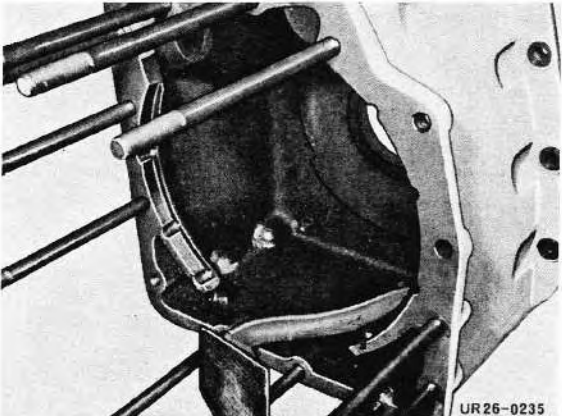
21 Remove main shaft.

22 Disassemble main shaft, see 6.1/1.



Valid for all Transmission Models

23 Using special tool No. 15 or No. 16, remove taper roller bearing from counter shaft, if necessary, also from input and main shaft.



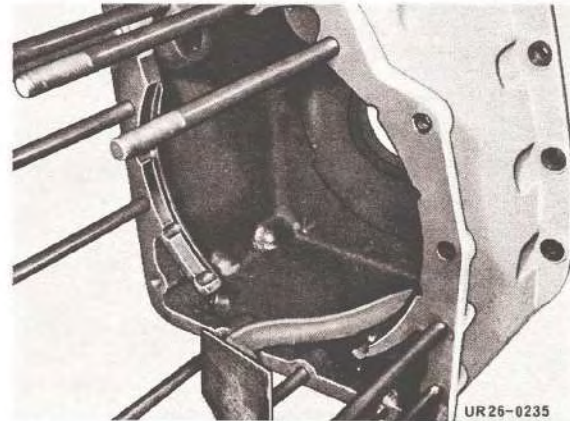
24 Remove oil intake pipe, together with oil filter.

Note: During each repair, remove and clean the filter on the oil intake pipes.

25 Clean all parts, check and renew if necessary.

Assembly of transmission

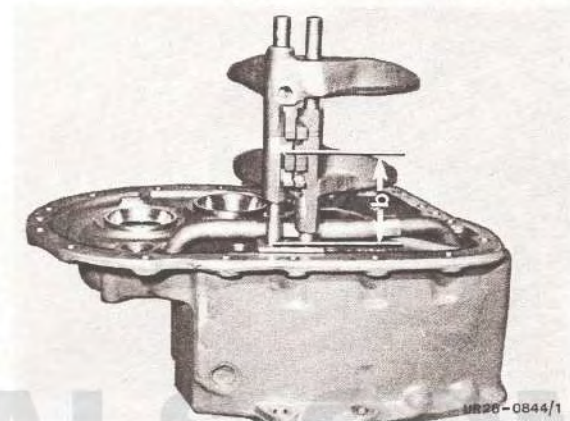
- 1 Clamp planetary casing in assembly stand.
- 2 Fit oil pipe with strainer.
- 3 Press outer races of taper roller bearings into epicyclic casing.
- 4 Rebuild selector shafts.



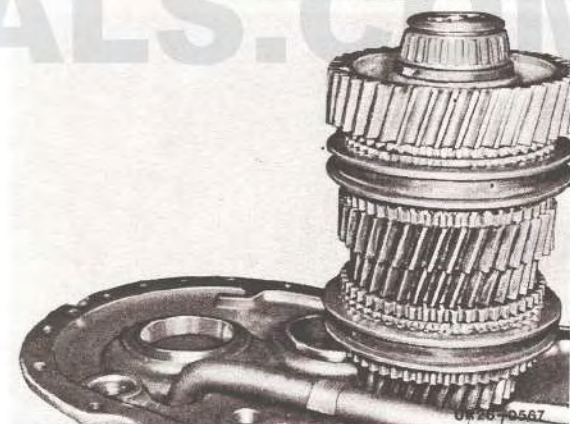
- 5 Fit and adjust selector shafts and guide plate. See 1.5/1 for adjustment.

Note: Make sure that selector forks are parallel.

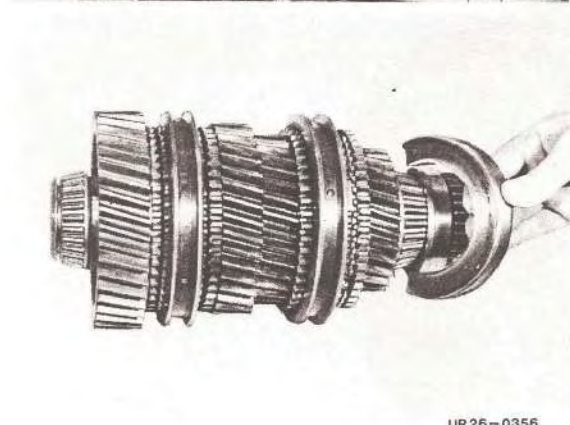
- 6 Remove selector shafts again.
- 7 Rebuild main shaft, see 6.1/2.
- 8 Rebuild primary shaft, see 5.1/1.



- 9 Fit main shaft, applicable to transmission model 717.901.

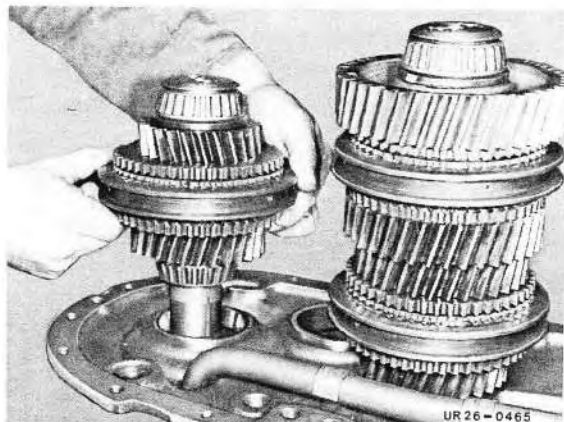


- 10 Attach special tool No. 8a and fit main shaft. Applicable to transmission model 717.900.



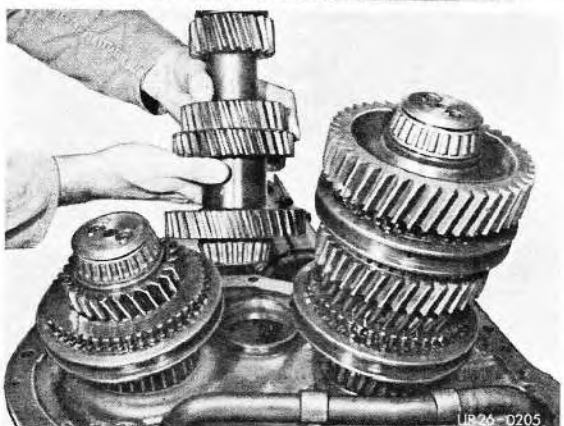
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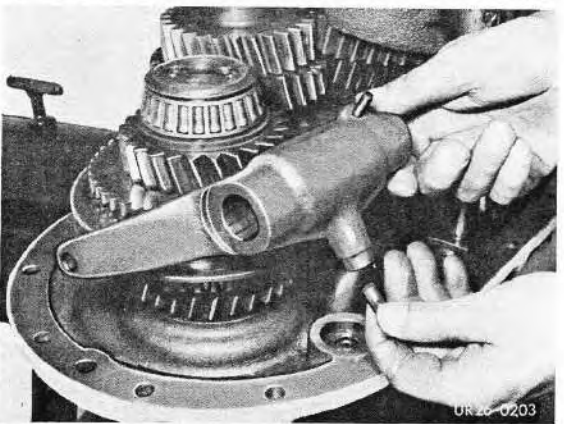


11 Fit primary shaft.

12 Press taper roller bearings on to layshaft.

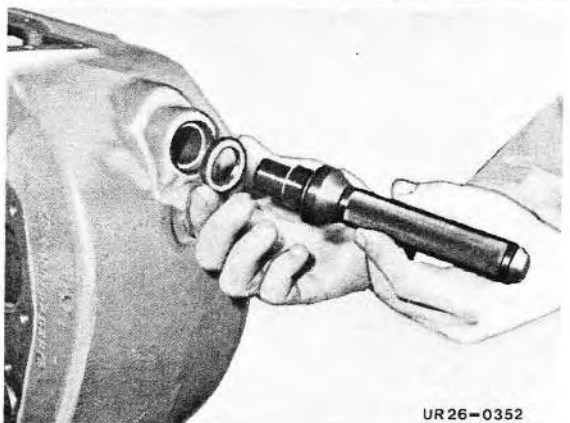


13 Tilt both shafts outwards and fit layshaft.



14 Fit forward/reverse selector fork with slide rings.

15 Fit guide plate, the necessary compensating shims and the selector shafts with the selector forks.



16 Coat shaft seal with grease No. 3 and press in with special tool No. 9.

17 Coat manual transmission housings with sealing compound No. 2 and bolt to planetary gearbox.

Referred: From chassis end No. 102 873 normal bolts are installed as standard in place of the locking bolts.

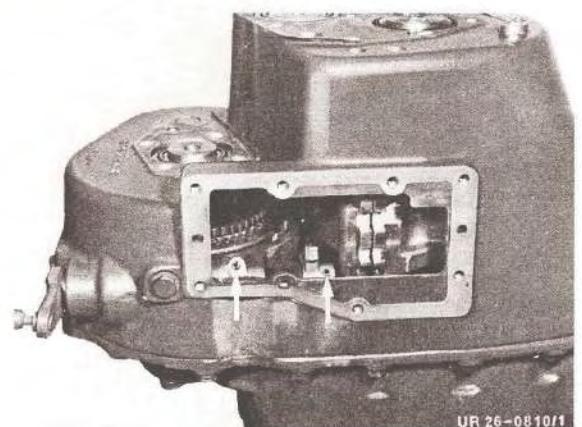
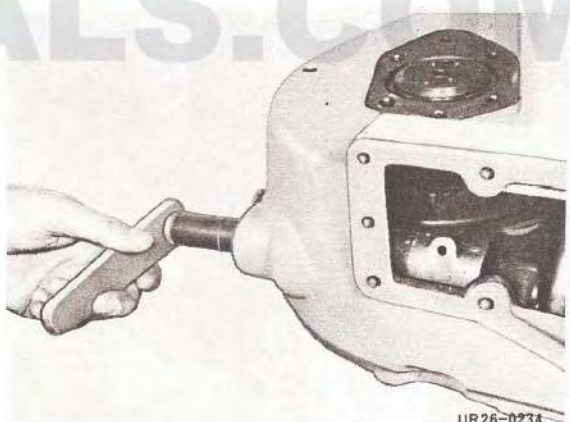
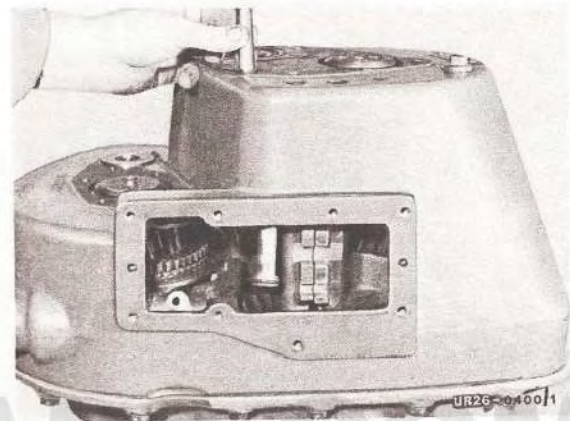
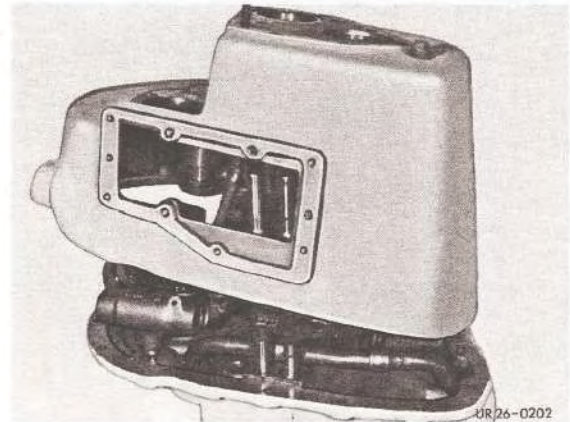
18 Press outer race of tapered roller bearing into manual transmission housing.

19 Coat oil connecting pipe with adhesive No. 4 and insert.

20 Install shifting shaft for forward or reverse gear and reverse shift lock in zero position so that the lever is positioned horizontally. Install dowel pin.

21 Secure shifting shaft and reverse shift lock.

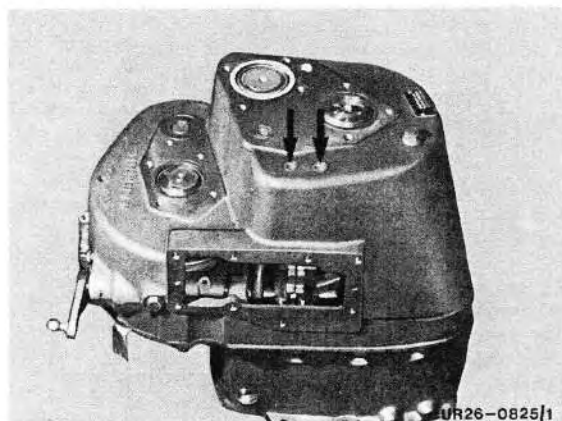
From transmission end No. 003 022, the reverse gear control lever lock is installed on request.



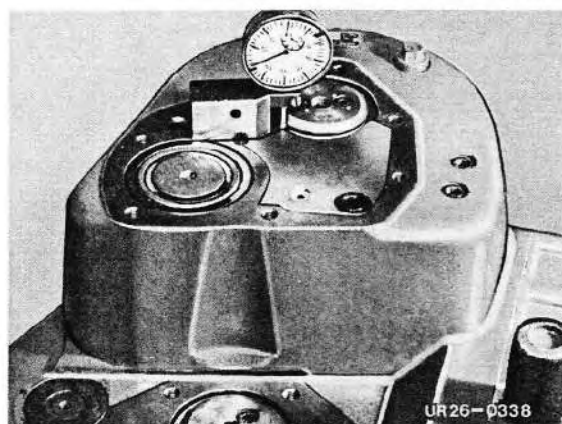
26.13 Assembly of Main Transmission

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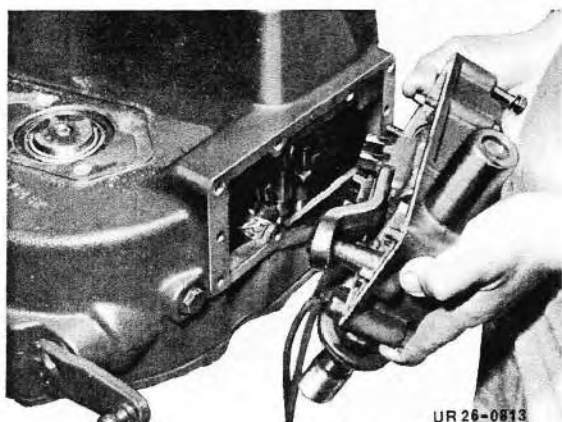


22 Tighten adjusting screws for the shift rods and coat with adhesive lacquer No. 6
For permissible tightening torques, see 1.7/1.



23 Place gasket in position and measure all bearings using special tool No. 13 and No. 18
For permissible play, see 1.5/1.

24 Place bearing end cover on input shaft, countershaft and main shaft and tighten.
For permissible tightening torques, see 1.7/1.



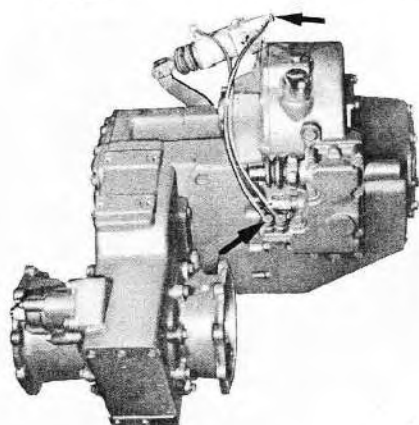
25 Install gear shift cover in neutral position and tighten. Shift through all gears once
For permissible tightening torques, see 1.7/1.

26 Turn gearbox housing through 180°.

27 Screw all remaining fastening bolts into planetary gearbox.

Note: From chassis end No. 102 873 normal bolts are installed as standard in place of locking bolts.

28 Install pneumatic cylinder with bracket.



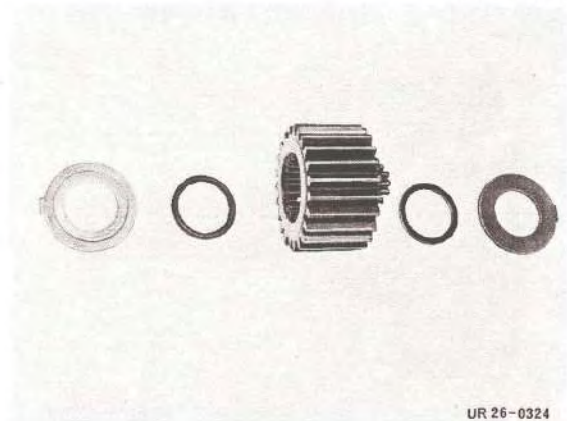
29 Connect delivery line crosswise.

Note: The fast group (5th to 8th speed) must be disengaged when the pneumatic cylinder is retracted.

Assembly of planetary gear

1 Insert bearing needles and thrust washers with lubricating grease No. 5, installing in planet pinion.

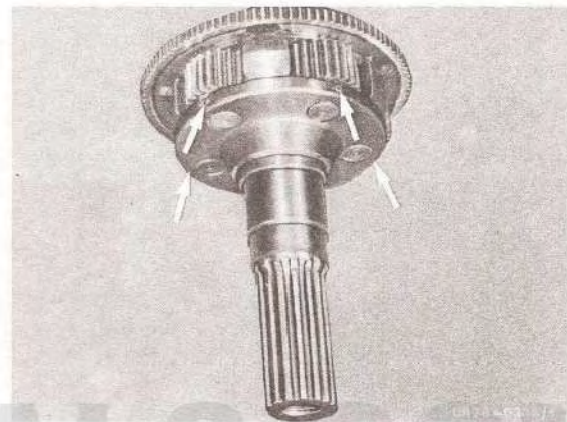
Note: Bearing needles must always be exchanged in sets (22 per set).



UR 26-0324

2 Slip planet pinions into planet-pinion carrier with shims, and install planet pinion shafts.

3 Screw planet-pinion shafts in position and secure with center punch.



UR 26-0371

4 Install and secure ring gear, applicable up to transmission No. 012 581.

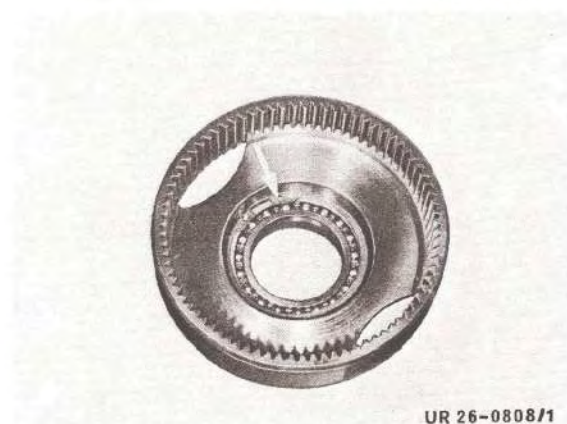


UR 26-0809

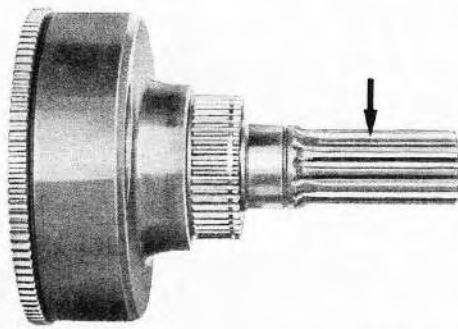
5 Install and secure deep-groove ball bearing.

Note: Deep-groove ball bearing has a certain amount of play for self-centering action of sun gear and planet pinions.

See 1.5/1 for permissible play.



UR 26-0808/1



UR 26-0807/1

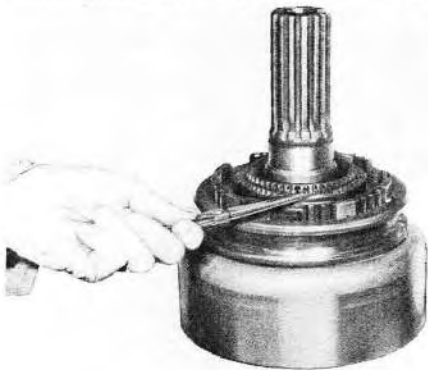
- 6 Install planet-pinion carrier in ring gear.



UR 26-Q386

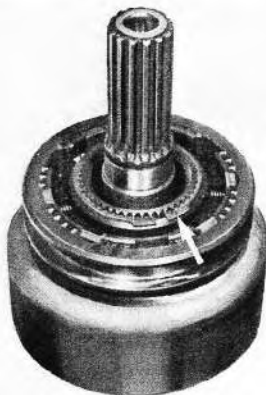
- 7 Install and secure synchronizing dome.

- 8 Assemble synchromesh member with dog and synchronizing cone, referring to 7.1/1.



UR 26-0814

- 9 Install synchromesh member and check bearing play.
See 1.5/1 for permissible play.



UR 26-0806/1

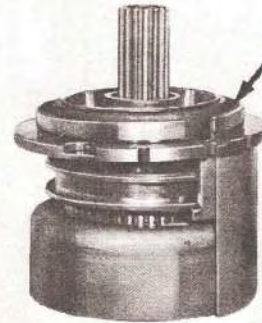
- 10 Secure synchromesh member.

- 11 Fit distance tube and compensating shim.



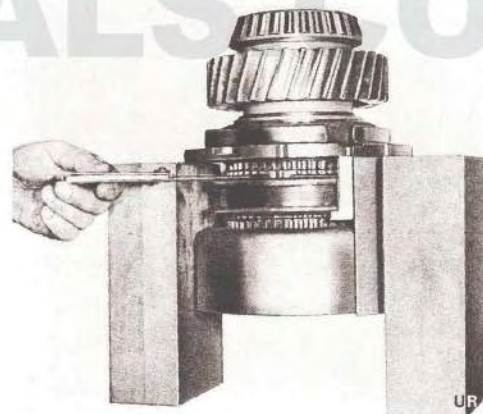
UR 26-0828

- 12 Fit synchro hub.
13 Put input gear on to planet carrier and fasten with a bolt.



UR 26-0804/1

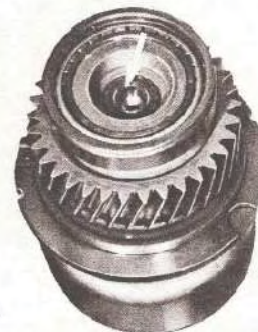
- 14 Support planet carrier and check synchromesh play.
Permissible synchromesh play, see 1.5/1.



UR 26-0815

- 15 Unscrew bolt, measure bearing cap and input gear and determine the compensating shims.
Permissible end float, see 1.5/1.

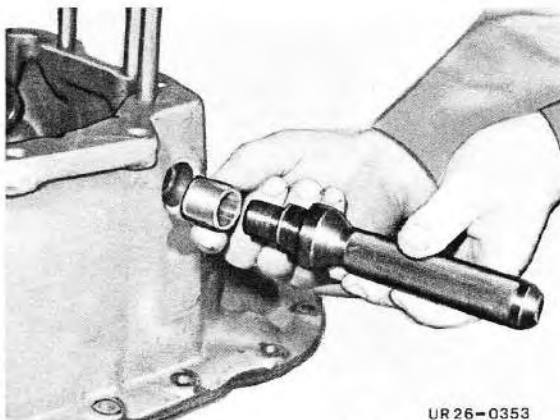
- 16 Remove input gear and fit selector fork.



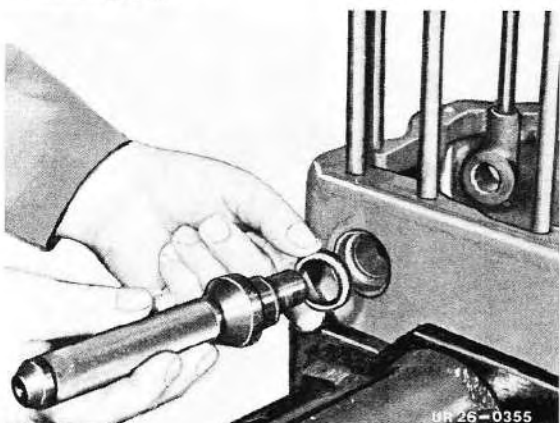
UR 26-0816/2

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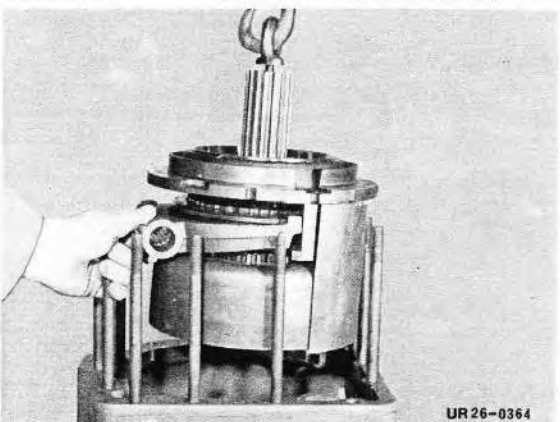
SA 35 977



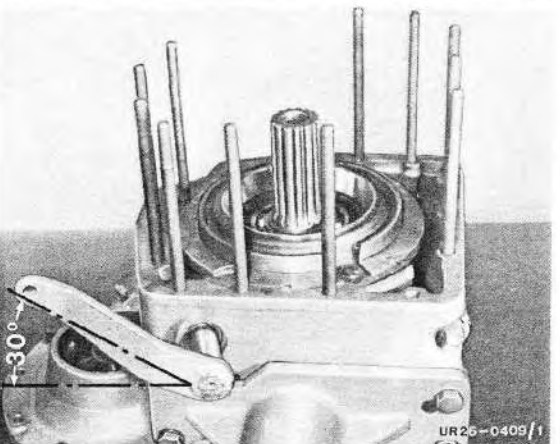
UR 26-0353



UR 26-0355



UR 26-0364



UR 26-0409/1

Installation of planetary gear

1 Check bearing bush in the selector shaft drilling, remove if necessary and fit new bush with special tool No. 9 and No. 10.

2 Coat shaft seal with grease No. 7 and fit with special tool No. 9.

3 Install planetary gearing with selector fork.

Note: Watch out for oil deflector plate.

4 Fit torsion bolts.

5 Coat shaft seal with grease No. 7 and press in with special tool No. 5.

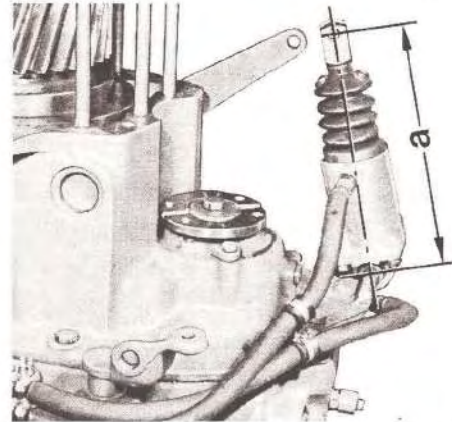
6 Fit the selector lever in such a way that the welded lever in its central position points upwards at about 30° to the horizontal.

7 Fit dowel pin to selector lever.

8 Adjust pneumatic cylinder.
For adjustment, see 1.5/1.

Adjustment

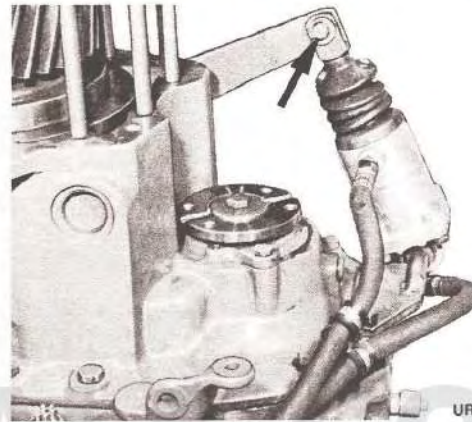
- Move piston rod to stop by pulling out.
- Set specified dimension by turning the clevis.
- Secure adjusted clevis by locking the nut.
- Pull gaiter over clevis.



UR26-1439/1

9 Secure shift lever to pneumatic cylinder.

Note: The fast group speeds 5 to 8 must be disengaged when the shift cylinder is retracted.



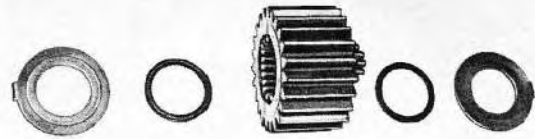
UR26-1440/1

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Assembly of Planetary Gear

1 Insert bearing needles and thrust washers packed with grease No. 5 and install in planet pinion.

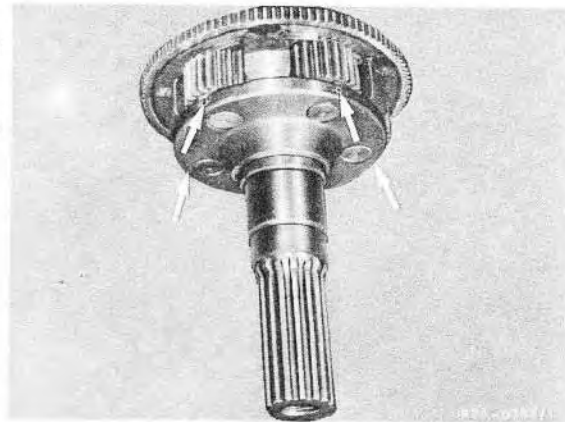
Note: Bearing needles must always be exchanged in sets (22 per set).



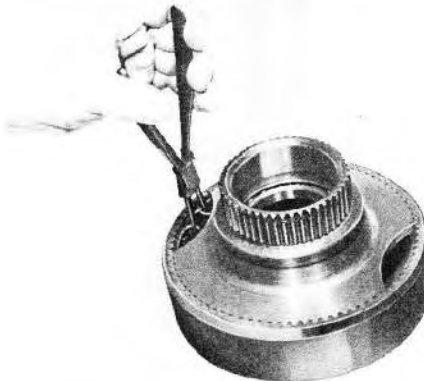
UR 26-0324

2 Fit planet pinions together with shims into planet pinion carrier and install planet pinion shaft.

3 Secure planet pinion shaft in position with center punched bolt.



4 Install ring gear and secure.
Valid to transmission No. 012 581.



UR 26-0809

5 Install deep-groove ball bearing and secure.

Note: Deep-groove ball bearing has a certain amount of play for self-centering of the sun gear and the planet pinions.
See 1.5/1 for permissible radial play.

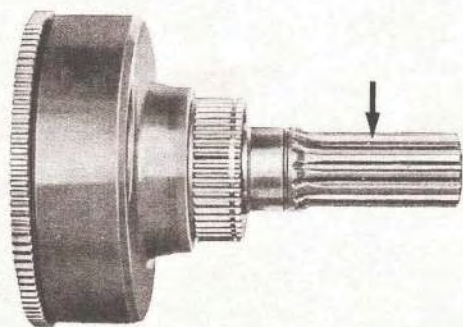


UR 26-0808/1

26.13 Assembly of Main Transmission

717.9
SA 35 977

- 6 Install planet pinion carrier in ring gear.



UR 26-0807/1

- 7 Install synchronizing dome and secure.

- 8 Assemble synchromesh member, see 7.1/1.



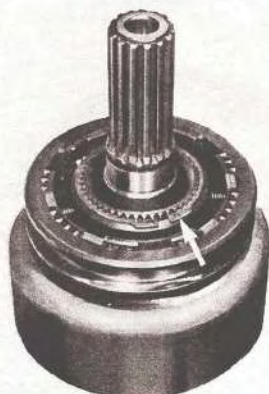
UR26-0388

- 9 Install synchromesh member, together with driver and sliding sleeve, check axial play.
See 1.5/1 for permissible axial play.



UR 26-0814

- 10 Secure synchromesh member.



UR 26-0806/1